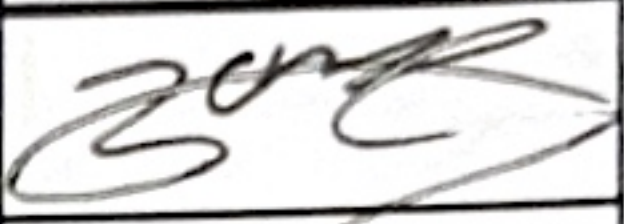
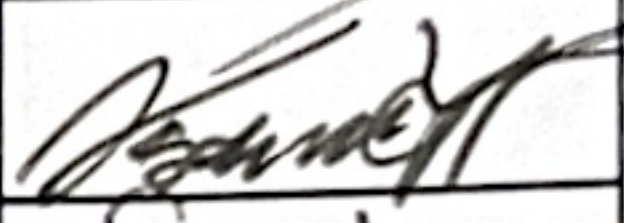
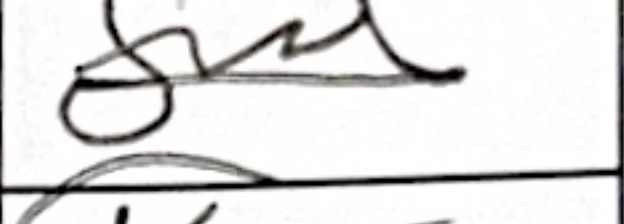
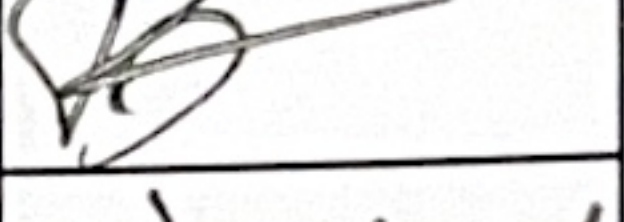
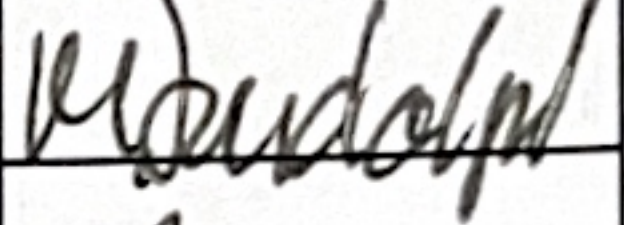


Session 1

Date: 26/03/2026

Name	Study Number	Present	Absent	Note	Signature
Jonas Jensen	s240324	x			
Andreas Jacobsen	s241123	x			
Sigurd Hestbech	s245534	x			
Tobias Nilsson	s233987	x			
Mads Rudolph	s246132	x			
Anders Falkesgaard	s245905	x			

Task carried out during the session:

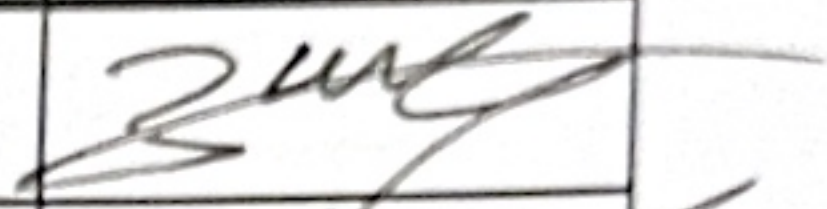
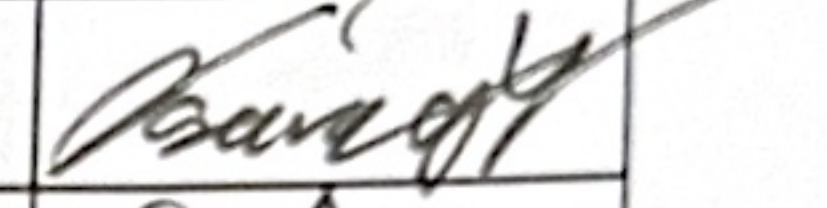
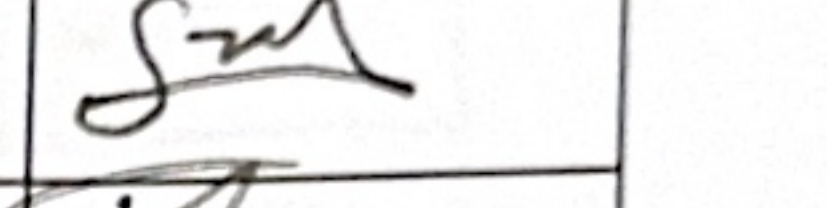
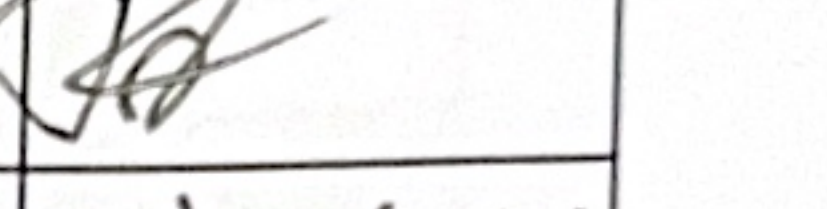
- *Brainstorming ideas*
We are pretty sure we will go with a smart thrash can that automatically seals and notifies the user.
- *Group Contract*
Read and signed by each member
- *Group Discord created*

Task to be done before next week:

- *Easter holiday*
- We will each think about possible challenges and design choices before next time
- Fill out and submit our project description (Andreas is lead, but we will coordinate through discord)

Session 2

Date: 09/04/2026

Name	Study Number	Present	Absent	Note	Signature
Jonas Jensen	s240324	x			
Andreas Jacobsen	s241123	x			
Sigurd Hestbech	s245534		x	Sick	
Tobias Nilsson	s233987	x			
Mads Rudolph	s246132	x			
Anders Falkesgaard	s245905		x	Sick	

Task carried out during the session:

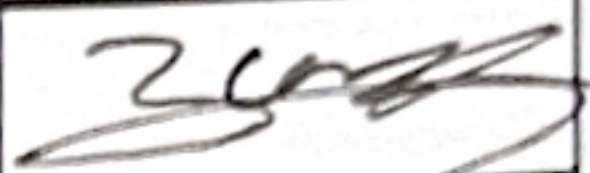
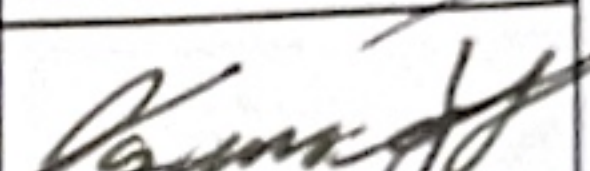
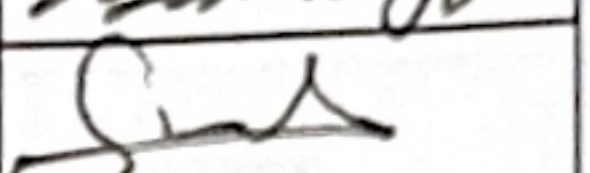
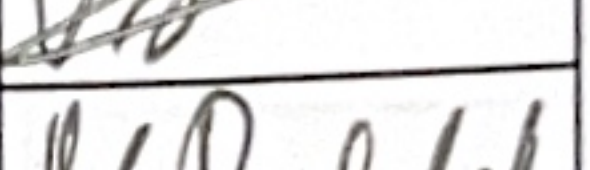
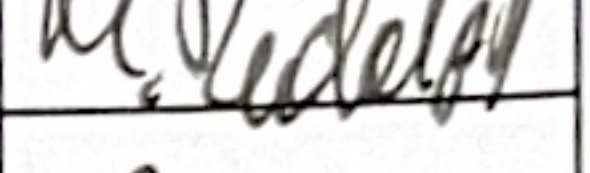
- Github created
- Discussion and final decision to go with the trash can project.
- Further brainstorming about the general design choice (Which type of bags, size, mechanism options, etc.)

Task to be done before next week:

- Create a state diagram
- Figure out how to think of ideas and maybe implement them
- Create a list of ideas
- Create a list of ideas
- Create a list of ideas

Session 3

Date: 16/04/2026

Name	Study Number	Present	Absent	Note	Signature
Jonas Jensen	s240324	x			
Andreas Jacobsen	s241123		x	Sick	
Sigurd Hestbech	s245534	x			
Tobias Nilsson	s233987	x			
Mads Rudolph	s246132	x			
Anders Falkesgaard	s245905	x			

Task carried out during the session:

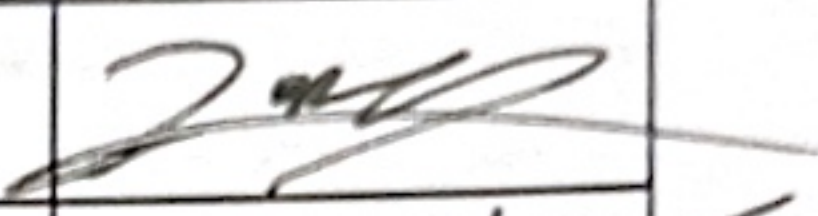
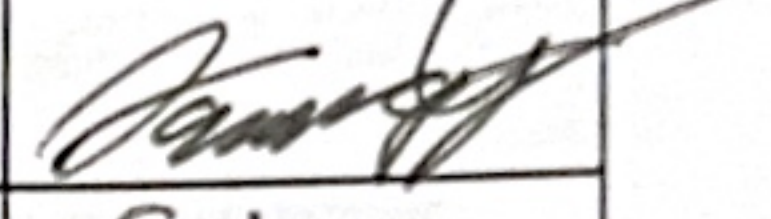
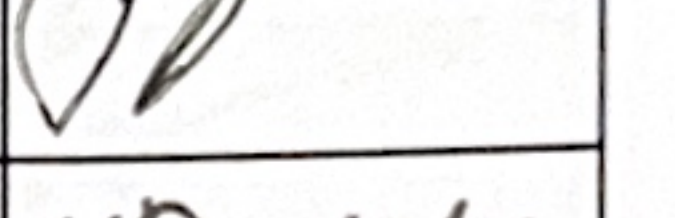
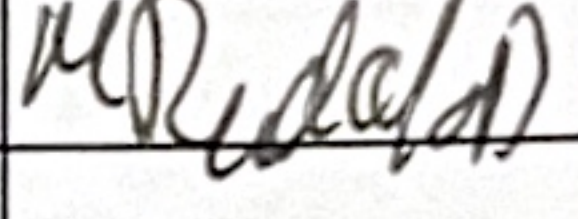
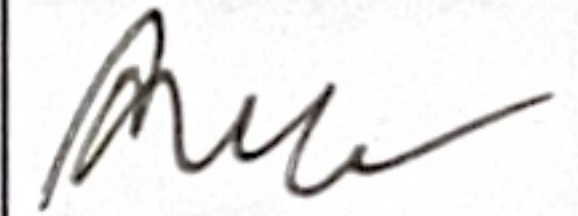
- Sigurd and Anders began designing the mechanism in CAD
- Jonas and Tobias began to examine and test different motor options and their compatibility with the sensors at our disposal.
- Mads began to design our code.
- Anders created a state diagram.

Task to be done before next week:

- *Sigurd*
Sigurd will continue to think of ideas and maybe implement into the next prototype.
- *Anders*
Print the first prototype before next weeks session.

Session 4 (Spontaneous)

Date: 23/04/2026

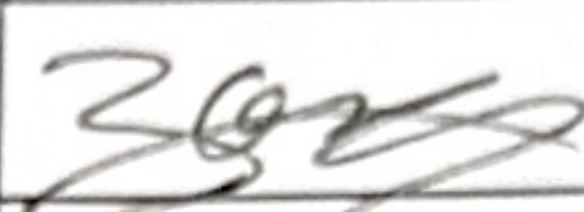
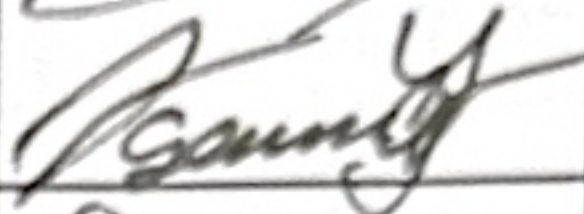
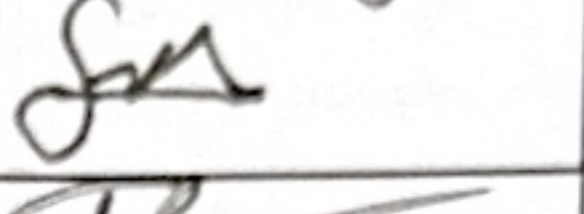
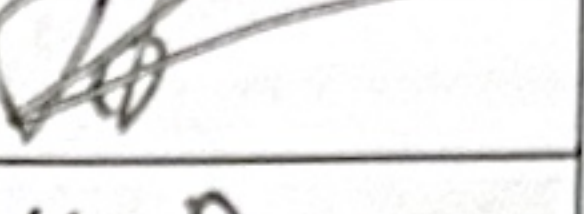
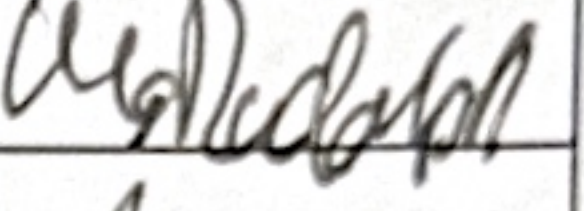
Name	Study Number	Present	Absent	Note	Signature
Jonas Jensen	s240324	x			
Andreas Jacobsen	s241123	x			
Sigurd Hestbech	s245534	x			
Tobias Nilsson	s233987	x			
Mads Rudolph	s246132		x	Sick	
Anders Falkesgaard	s245905	x			

Task carried out during the session:

- Sigurd continues with CAD and makes adjustments based on our learnings with the first prototype.
- Jonas, Tobias and Andreas is beginning to implement the different motor options into the first prototype, and test with the draft of our code.
- Anders have created and begun writing an Overleaf document for the report as well as streamlining the log book proces.

Session 5 (Spontaneous)

Date: 26/04/2026

Name	Study Number	Present	Absent	Note	Signature
Jonas Jensen	s240324	x			
Andreas Jacobsen	s241123	x			
Sigurd Hestbech	s245534	x			
Tobias Nilsson	s233987		x		
Mads Rudolph	s246132		x		
Anders Falkesgaard	s245905		x		

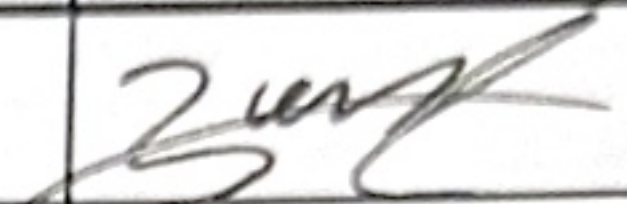
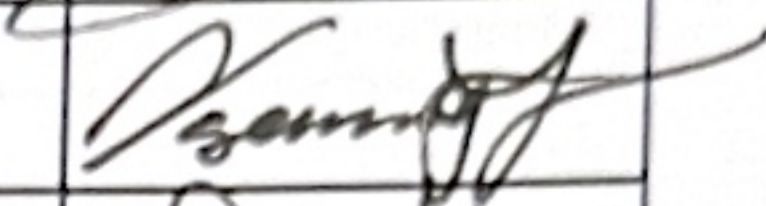
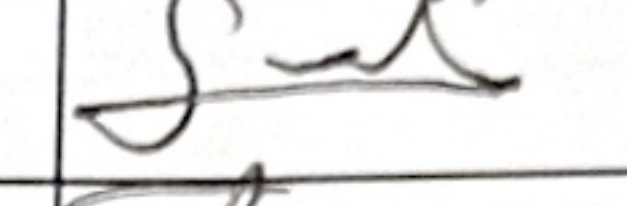
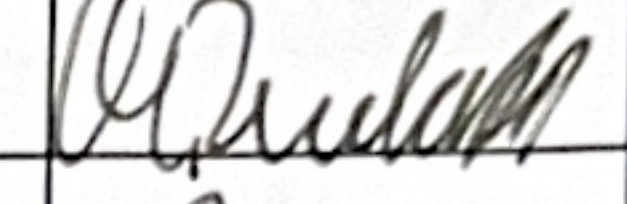
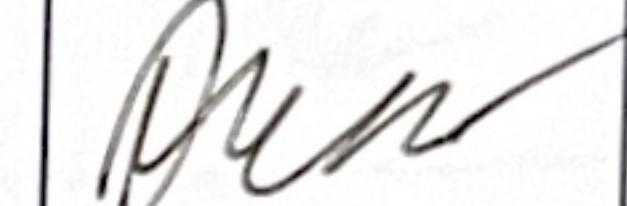
Jonas, Andreas and Sigurd met up spontaneously

Task carried out during the session:

Jonas, Andreas and Sigurd implemented the state machine and made some new CAD-revisions.

Session 6

Date: 30/04/2026

Name	Study Number	Present	Absent	Note	Signature
Jonas Jensen	s240324	x			
Andreas Jacobsen	s241123	x			
Sigurd Hestbech	s245534	x			
Tobias Nilsson	s233987	x			
Mads Rudolph	s246132	x			
Anders Falkesgaard	s245905	x			

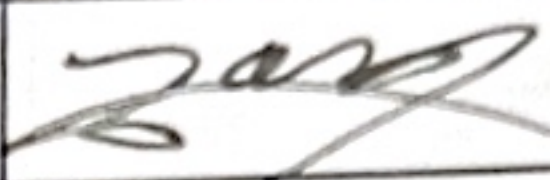
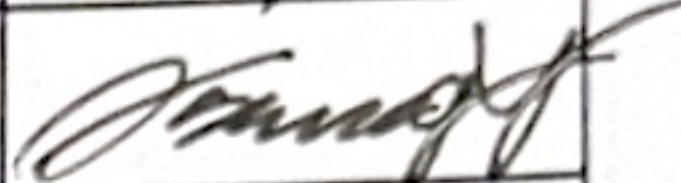
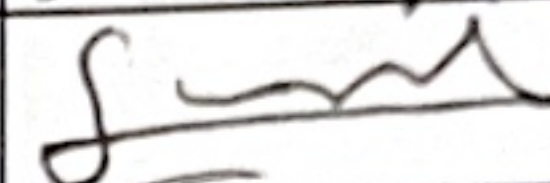
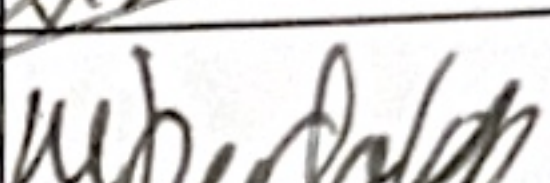
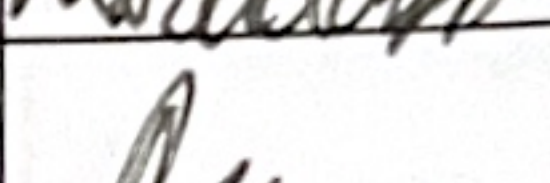
We started the day in Lyngby where we heard the presentation and had a status meeting with our mentor. Afterwards we drove to Ballerup, where we have access to a coworking space as well as two 3D-printers.

Task carried out during the session:

- Mads integrated ThingSpeak and IFTTT into the firmware: the bin now pushes bag count, distance, state and bag-full events to the cloud, and reads bag-count updates from the phone via Field 5.
- Set up the ThingSpeak channel with ThingHTTP entries and React rules so a "bag full" or "low bags" event triggers a push notification on the phone via IFTTT.
- Updated the README to reflect the current IoT pipeline and the new dashboard.
- We worked on optimizing the code (adjusted stepper distance, code cleanup) and troubleshooting the closing mechanism.
- Added a MOSFET on the stepper-driver power line to make it more efficient and reduce idle current.
- Tobias started the custom PCB-design

Session 7

Date: 01/05/2026

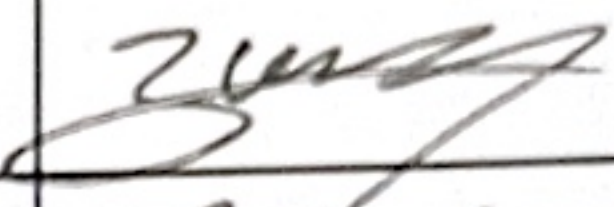
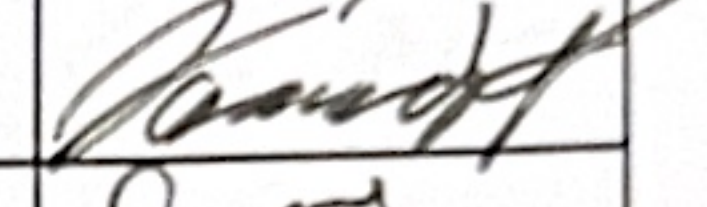
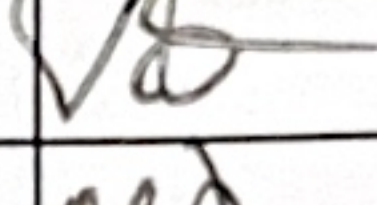
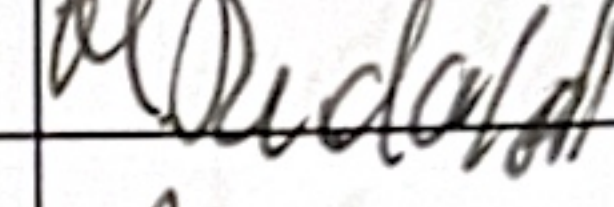
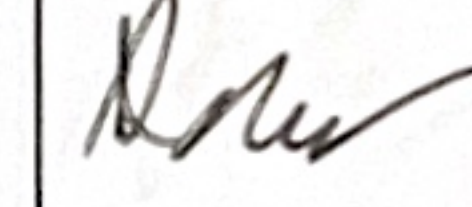
Name	Study Number	Present	Absent	Note	Signature
Jonas Jensen	s240324	x			
Andreas Jacobsen	s241123	x			
Sigurd Hestbech	s245534	x			
Tobias Nilsson	s233987	x			
Mads Rudolph	s246132	x			
Anders Falkesgaard	s245905	x			

Task carried out during the session:

- The mechanism is having trouble to get a proper grip on the strings of the bag, Sigurd have changed the rolls to accommodate two O-rings, we will test tomorrow after the print is done.
- The MOSFET added in last session was removed because it conflicted with the TX pin and broke Serial logging. Instead the stepper coil pins are now pulled LOW after each move, which gives the same idle-current saving without an extra component.
- Mads redesigned the web dashboard: translated the UI from Danish to English, added a temperature card with a thermometer SVG, and added a status pill that flips to OFFLINE when the ESP hasn't pushed data for >90 s.

Session 8

Date: 02/05/2026

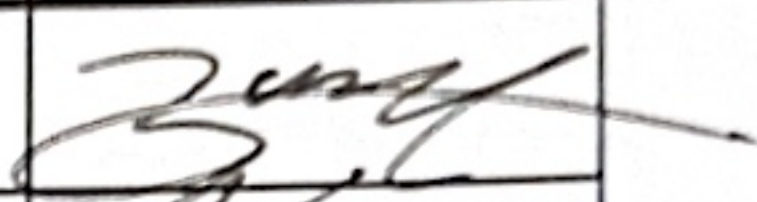
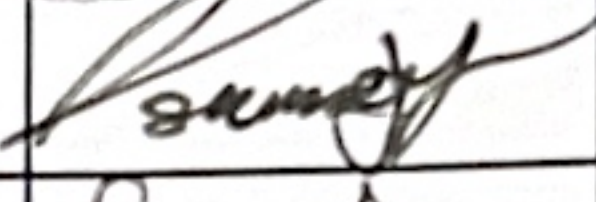
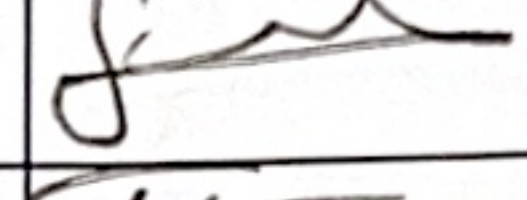
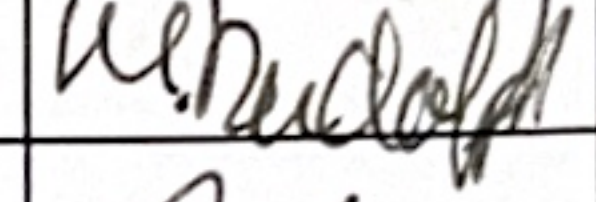
Name	Study Number	Present	Absent	Note	Signature
Jonas Jensen	s240324	x			
Andreas Jacobsen	s241123	x			
Sigurd Hestbech	s245534	x			
Tobias Nilsson	s233987	x			
Mads Rudolph	s246132	x			
Anders Falkesgaard	s245905	x			

Task carried out during the session:

- The bag is not closing reliably, so we will install a second motor. We will therefore go from a single-motor design to a dual-motor setup.
- Sigurd Is redesigning the gears and housing to make space for the extra motor, Anders will print it before next session (tomorrow)
- Anders, Jonas and Andreas added an SSD1306 OLED display and a temperature sensor to the bin. Started with a DHT11 (temp + humidity) but switched to a simpler LM35 (temperature only), mainly due to the requirement for an analog sensor.
- Mads updated the ThingSpeak push to send the value on Field 6.

Session 9

Date: 03/05/2026

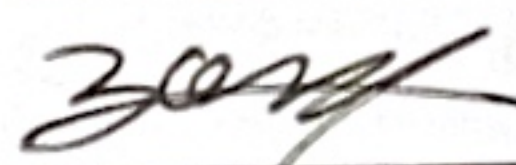
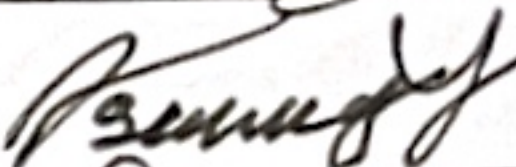
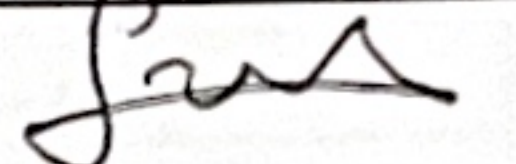
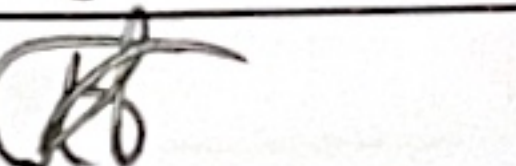
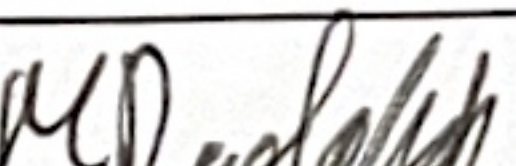
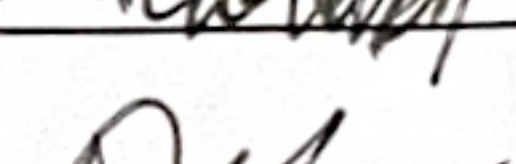
Name	Study Number	Present	Absent	Note	Signature
Jonas Jensen	s240324	x			
Andreas Jacobsen	s241123	x			
Sigurd Hestbech	s245534	x		Work	
Tobias Nilsson	s233987	x			
Mads Rudolph	s246132	x			
Anders Falkesgaard	s245905	x			

Task carried out during the session:

- Started setting up the Poster with the different relevant sections (State, Schematic, Photo, ThingsSpeak)
- Anders calibrated the LM35 ADC reference voltage to 2.667 V via an external thermometer and datasheet.
- Andreas and Anders added an IIR filter (10-sample average + 30/70 weighted blend) to smooth the temperature readings. It fluctuated quite a bit due to imprecise ADC on the ESP8266, it is also very affected by the noise from the Wifi-module.
- Jonas fixed the stepper pinout (moved IN4 from D1 to D0 so the OLED can use the I2C SCL line) and adjusted the closing distance in EMPTY_ME.
- Mads wrote an explainer document (How the web dashboard works.md) describing the ESP ↔ ThingSpeak ↔ dashboard flow for the rest of the team and the report.
- Tobias' first PCB is finished, but lacks full functionality, he will redesign and go for a v2
- Sigurd made more revisions to cad and prepared more files for printing

Session 10

Date: 04/05/2026

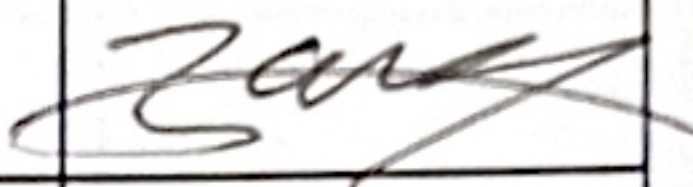
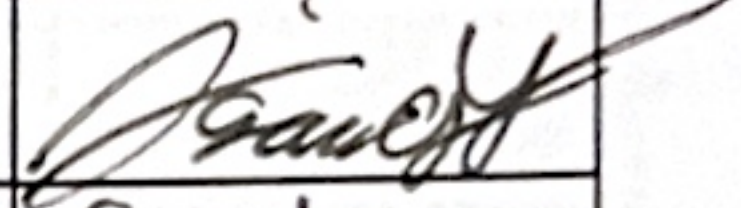
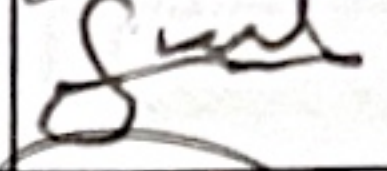
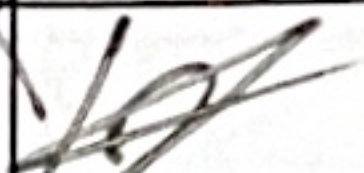
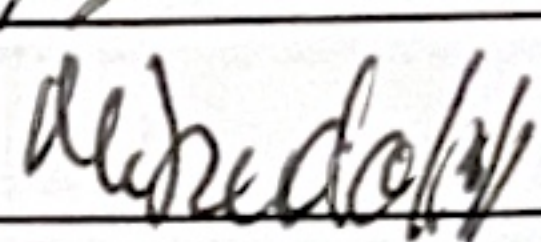
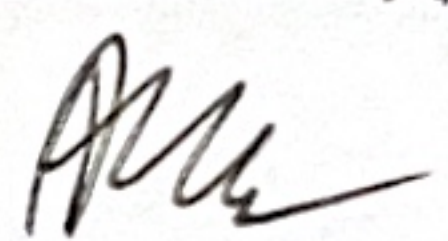
Name	Study Number	Present	Absent	Note	Signature
Jonas Jensen	s240324	x			
Andreas Jacobsen	s241123	x			
Sigurd Hestbech	s245534		x	Work	
Tobias Nilsson	s233987	x			
Mads Rudolph	s246132	x			
Anders Falkesgaard	s245905	x			

Task carried out during the session:

- Jonas and Andreas is working on optimizing the code.
The light-sleep mode had a conflict with the I2C during startup of the chip. So we discussed depending on Modem Sleep only, but ultimately agreed on trying to get the light-sleep to work. Which they succeeded with (Boot order).
- Anders began the power calculations.
- We continued setting up the poster. The photo has been scrapped (The prototype will stand next to the poster). Made space for the PCB-schematic as well. Began designing the space for the power calculations in the middle of the poster.
- Built and deployed a small web dashboard at <https://dtutrash.netlify.app/> for viewing live bag count, distance and state, with an input field to set the bag count remotely.
- Tobias made final revisions on v2 of the PCB and began the printing, it should be done tomorrow.
- Sometimes the bag strings are getting caught in the gears, Sigurd designed some spacers from home

Session 11

Date: 05/05/2026

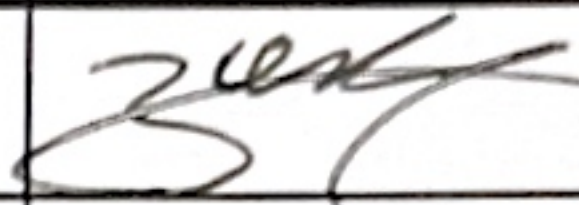
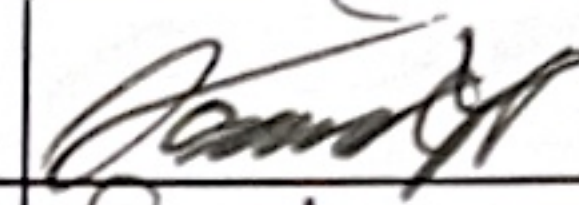
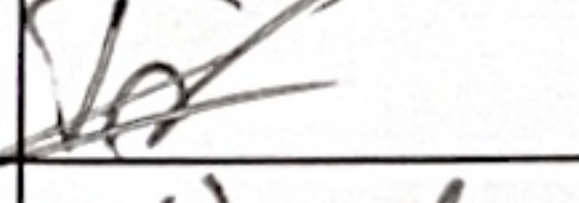
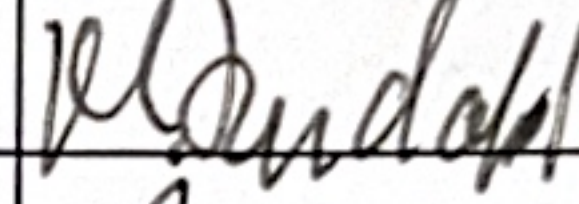
Name	Study Number	Present	Absent	Note	Signature
Jonas Jensen	s240324	x			
Andreas Jacobsen	s241123	x			
Sigurd Hestbech	s245534	x			
Tobias Nilsson	s233987	x			
Mads Rudolph	s246132	x			
Anders Falkesgaard	s245905	x			

Task carried out during the session:

- Even with the spacers, the trash can still has problems closing reliably, Sigurd has made room for more O-rings on the wheels, we will print and test tomorrow. Hopefully final revision.
- Anders finalized the power calculations
- Andreas, Jonas, Sigurd, Tobias was combining the final parts (minus the new wheels) and mounted the PCB to the trash can. (The lid is opening the wrong way, but we determined that it was a minor problem easily explained at the poster presentation.)
- Spent the rest of the day testing the trash can and found a loose VCC wire to one of the motor drivers, which means that one motor should have to drive both motors + the wheels and bag. Tobias found and fixed the problem, but we will implement the new wheels as well tomorrow.
- Mads designed a flow-diagram for the Thingspeak part of the poster

Session 12

Date: 06/05/2026

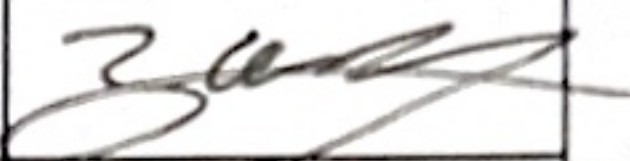
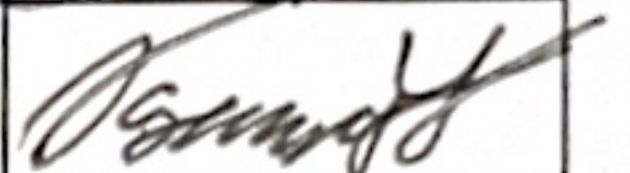
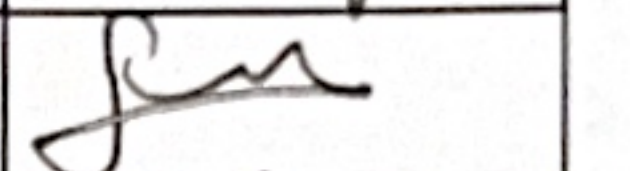
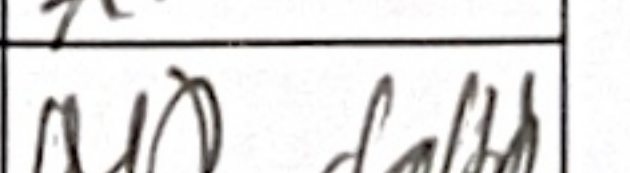
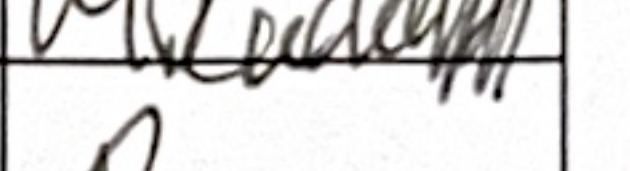
Name	Study Number	Present	Absent	Note	Signature
Jonas Jensen	s240324	x			
Andreas Jacobsen	s241123	x			
Sigurd Hestbech	s245534	x			
Tobias Nilsson	s233987	x			
Mads Rudolph	s246132	x			
Anders Falkesgaard	s245905	x			

Task carried out during the session:

- Final assembly, everything works as intended.
- Anders made final touches on the poster + print at Lyngby after lecture
- We made a video on the final product
- All practiced the poster presentation in front of each other

Session 13

Date: 12/05/2026

Name	Study Number	Present	Absent	Note	Signature
Jonas Jensen	s240324	x			
Andreas Jacobsen	s241123	x			
Sigurd Hestbech	s245534	x			
Tobias Nilsson	s233987	x			
Mads Rudolph	s246132	x			
Anders Falkesgaard	s245905	x			

Task carried out during the session:

- Everyone has filled out parts of the report from home individually. today we meet up to combine and agree on the final report.
- We filled out the group evaluation at the end of the day